A Fast Iterative Shrinkage-Thresholding Algorithm for Linear Inverse Problems

Abstract:

This project aims to restore clear and high-quality images from blurred and noisy versions using two optimization algorithms: the Iterative Shrinkage-Thresholding Algorithm (ISTA) and its faster variant, the Fast Iterative Shrinkage-Thresholding Algorithm (FISTA). Based on the work of Beck and Teboulle, the problem is treated as a linear inverse problem where a degraded image is reconstructed by balancing two goals — making the result match the observed image and enforcing a level of sparsity to reduce noise. In our approach, a clean grayscale image is first taken and deliberately degraded by applying a blur filter and adding artificial noise to simulate real-world issues such as camera shake or sensor errors. Both ISTA and FISTA are then implemented in Python to recover the original image. FISTA improves over ISTA by using an acceleration technique that achieves the same or better quality in fewer iterations, making it faster and more efficient. The project outputs include side-by-side visual comparisons of the restored images, plots showing how quickly each method converges, and quality measures such as image sharpness and similarity scores. Results clearly show that FISTA can produce better reconstructions in less time, making it a strong choice for image restoration tasks where speed and quality are equally important.

Applications:

1. **Medical Imaging** — Enhancing blurred or noisy MRI, CT, or X-ray scans for better diagnosis.
2. **Astronomy** - Restoring astronomical photos affected by atmospheric distortion.
3. **Forensics & Surveillance** — Recovering details from low-quality or blurred security footage.
4. **Photography** — Reducing motion blur and noise in consumer and professional camera images.